Generative AI Truth, and Rights challenges: Ethical Challenges in the Age of Synthetic Media

Exploring the impact and responsibilities of AI-generated content

What is Generative AI?



Definition of Generative AI

Generative AI creates new content, including text, images, audio, and video, using AI systems.

Advanced Machine Learning Models

Generative AI relies on models like transformers and diffusion to produce high-quality outputs, and large training datasets

Examples of Generative AI

Examples include large language models, image generators, and video synthesis tools.

Industry Transformations

Generative AI is transforming the marketing, entertainment, and education industries globally.

Capabilities & Use Cases



Diverse Business Applications

Generative AI supports content creation, personalized marketing, chatbots, and product design.

Efficiency and Innovation Benefits

AI capabilities boost efficiency and foster innovation across multiple business functions.

Societal Risks and Challenges

Generative AI can lead to misinformation, trust erosion, intellectual property issues, and ethical dilemmas.

Critical Thinking on Deployment

Encourages responsible and thoughtful use of AI to mitigate negative impacts.

Generative AI, Truth, and Intellectual Property



Introduction: The Double-Edged Sword of Generative AI

- What is Generative AI?
- Key capabilities: text, image, audio, and video generation
- Benefits for business and society
- Emerging ethical and legal challenges
  - Misinformation
  - Hallucinations
  - Deepfakes
  - Intellectual Property

Generative AI, Truth, and Intellectual Property



Misinformation: When AI Scales Falsehoods

Key Concepts:

- Misinformation vs. disinformation
- AI-generated fake news
- Synthetic content at scale

Key Risks:

Public confusion, Erosion of trust, Manipulation of public opinion, Business reputation damage

Generative AI, Truth, and Intellectual Property



Hallucinations: When AI Invents Facts

Key Concepts:

- Definition of hallucinations
- Why large language models hallucinate

Key Risks:

Fabricated citations and references; Incorrect professional advice; Poor decision-making

Business Implications:

Healthcare; Legal services; Education; Corporate communication

Generative AI, Truth, and Intellectual Property



Deepfakes: The Authenticity Crisis

Key Concepts:

- AI-generated synthetic audio, images, and videos
- Voice cloning and impersonation

Key Risks:

Fraud and social engineering; Reputational harm; Political manipulation; Loss of trust in digital evidence

Examples:

Fake CEO voice scams; Manipulated political videos

Generative AI, Truth, and Intellectual Property



Intellectual Property Challenges

Key Concepts:

- Copyright
- Ownership
- Authorship
- Fair use and training data

Key Risks:

Use of copyrighted material in model training; Ownership of AI-generated outputs; Style imitation and creative appropriation; Protection of creators' rights

Business Implications:

Copyright litigation; Licensing challenges; Regulatory uncertainty

Generative AI, Truth, and Intellectual Property



Conclusion: The Core Challenge of Generative AI

Truth Problems

- Misinformation
- Hallucinations
- Deepfakes

Rights Problems

Generative AI creates a dual challenge: it can undermine both the truthfulness of information and the ownership of creative work. Misinformation, hallucinations, and deepfakes threaten trust, while intellectual-property disputes challenge traditional notions of authorship, ownership, and creator rights.

LET US DIG DEEPER

Summary Table

Issue	*Core Problem*	*Main Consequence*
Misinformation	AI generates false content at scale	Trust erosion and social manipulation
Hallucinations	AI fabricates facts or sources	Poor decisions and reliability problems
Deepfakes	AI creates realistic fake media	Fraud, impersonation, reputational harm
Intellectual Property	Unclear ownership and copyright issues	Legal disputes and creator-rights concerns

Misinformation Defined

MISINFORMATION VS DISINFORMATION VS MALINFORMATION

MISINFORMATION	DISINFORMATION	MALINFORMATION
FALSE	FALSE	TRUE
NO INTENT TO HARM	INTENT TO DECEIVE	INTENT TO HARM
Sharing a rumor you think is true	Fabricating a fake news story	Leaking private emails

Definition of Misinformation

Misinformation is **false or misleading** information spread without intent to harm.

Types of problematic "information"

Includes disinformation, malinformation, and general misinformation with distinct motives.

Role of Generative AI

Generative AI creates realistic but false content, increasing misinformation spread rapidly – GAI Amplification

Impact on Digital Platforms

Social media amplifies misinformation created by AI, significantly influencing public perception.

Mechanisms of Misinformation



Deepfake Technology

Deepfakes manipulate video and audio to create realistic but false content that deceives viewers.

Synthetic Text Generation

AI-generated synthetic text can produce fake news articles and impersonate individuals online.

Social Media Bots

Automated bots spread misinformation rapidly across social media platforms, influencing public opinion.

Impacts on Trust

These mechanisms **erode public trust and distort public discourse** with misleading information.

Case Study: ChatGPT & Fake News



Have you encountered AI-generated misinformation?

AI-Generated Misinformation

ChatGPT and similar models can produce fake news and political misinformation, posing significant risks.

Ethical Challenges

Controlling AI outputs raises ethical questions about responsibility and accountability in AI use.

Responsibility Debate

Discussion focuses on whether developers, users, or platforms should be held accountable for misuse.

Preventive Safeguards

Implementing safeguards is crucial to prevent AI misuse and protect information integrity.

The New York Times

The People Onscreen Are Fake. The Disinformation Is Real.

Share full article · 455




By Adam Satariano and Paul Mozur
Adam Satariano, based in London, and Paul Mozur, based in Seoul, are tech correspondents who report internationally about online disinformation.

Feb. 7, 2023
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Sign up for the On Tech newsletter. Get our best tech reporting from the week. Get it sent to your inbox.

In one video, a news anchor with perfectly combed dark hair and a stubbly beard outlined what he saw as the United States' shameful lack of action against gun violence.







Case Study: ChatGPT & Fake News



Have you encountered AI-generated misinformation?

Example

Researchers showed that generative AI could produce realistic news reports claiming a politician had resigned or a company had committed fraud, even though the events never occurred. Once shared on social media, such content could influence public opinion before corrections are issued.

Discussion Questions

Should AI companies be responsible for preventing fake-news generation?

Should AI-generated content be clearly labelled?

Who is accountable when AI-generated misinformation causes harm?

Mitigation Strategies & Future Outlook

Detection & Moderation Tools



AI Watermarking

AI watermarking helps detect manipulated content by embedding invisible markers for verification.

Content Provenance Standards

Standards like C2PA ensure content authenticity by tracking origin and edits consistently.

Automated Fact-Checking

Automated systems rapidly verify factual claims to reduce misinformation spread.

Challenges of Scaling

Scaling tools needs balancing accuracy with speed and adapting to evolving misinformation tactics.



Business Strategy & Risk Management

Risk Assessment Frameworks

Implement frameworks to identify and evaluate AI-related risks effectively. This helps in making informed decisions.

Crisis Response Planning

Develop plans to respond swiftly to reputational or legal challenges caused by AI issues. Preparedness ensures resilience.

Communication Strategies

Establish transparent communication to maintain stakeholder trust and ethical leadership during AI risk events.

Future Trends



Multimodal Misinformation

Future AI will create misinformation using multiple media types, increasing complexity.

AI-Generated Lobbying

AI tools will automate and amplify lobbying efforts influencing policy and business.

Synthetic Propaganda

Synthetic media will be used to produce persuasive propaganda impacting society and politics.

Long-Term AI Impact

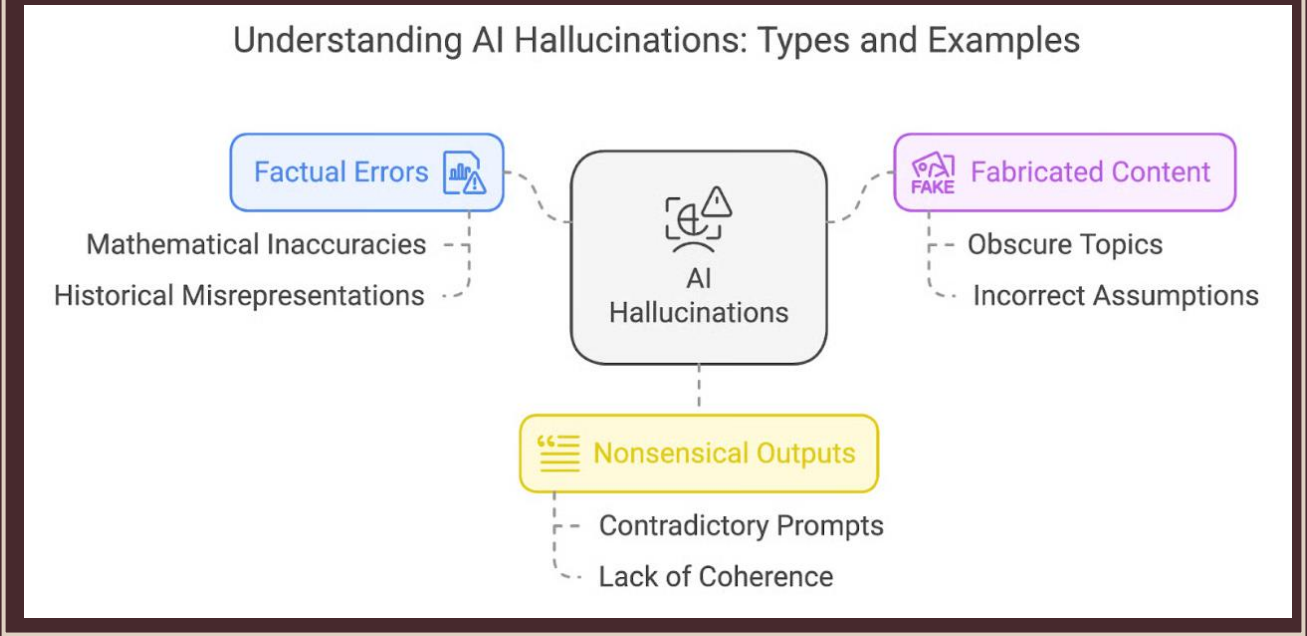
Understanding these trends helps anticipate future societal and business challenges.

The greatest long-term danger of AI misinformation is not simply that people will believe false information. It is that people may no longer know what information, evidence, or institutions can be trusted at all. This threatens the foundations of democratic governance, markets, science, and social cooperation

AI Hallucination: a type of AI misinformation



- An AI hallucination refers to an output generated by an AI model that deviates from reality or lacks a factual basis.
- In simpler terms, this is an instance where the model provides an incorrect answer, fabricates a story, or produces output that doesn't make sense. Hallucinations can vary in severity, ranging from a minor factual mistake to an entirely fabricated claim.



AI Hallucination: a type of AI misinformation



Why Do Hallucinations Occur?

- AI predicts the most likely next words rather than verifying facts.
- Models may lack relevant information.
- Ambiguous or complex prompts increase the risk of errors.

Examples

- Inventing academic articles or references
- Creating false statistics
- Misquoting people
- Fabricating legal cases or medical advice

Risks

- Poor business decisions
- Misinformation and loss of trust
- Legal, financial, and reputational harm

AI ETHICS CASE STUDY: META'S GALACTICA MODEL

EXPLORING ETHICAL CONSIDERATIONS IN ADVANCED AI SYSTEMS

OVERVIEW OF GALACTICA

Launch and Purpose

Galactica launched in November 2022 to organize and summarize scientific knowledge effectively.

Extensive Training Data

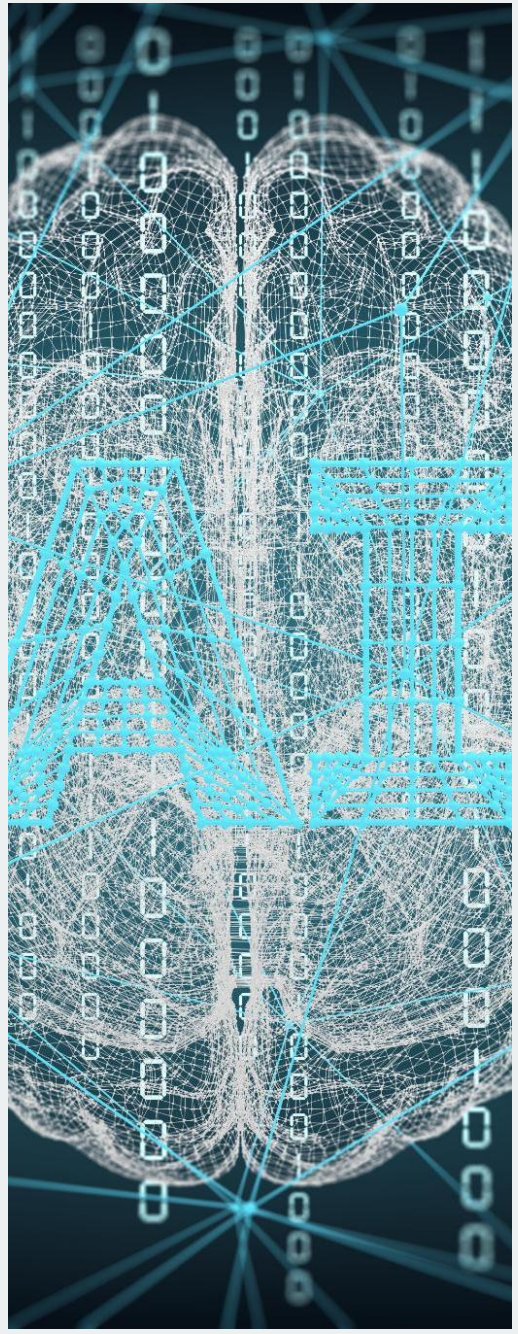
Trained on 48 million scientific papers, textbooks, and web sources like Wikipedia.

Versatile Capabilities

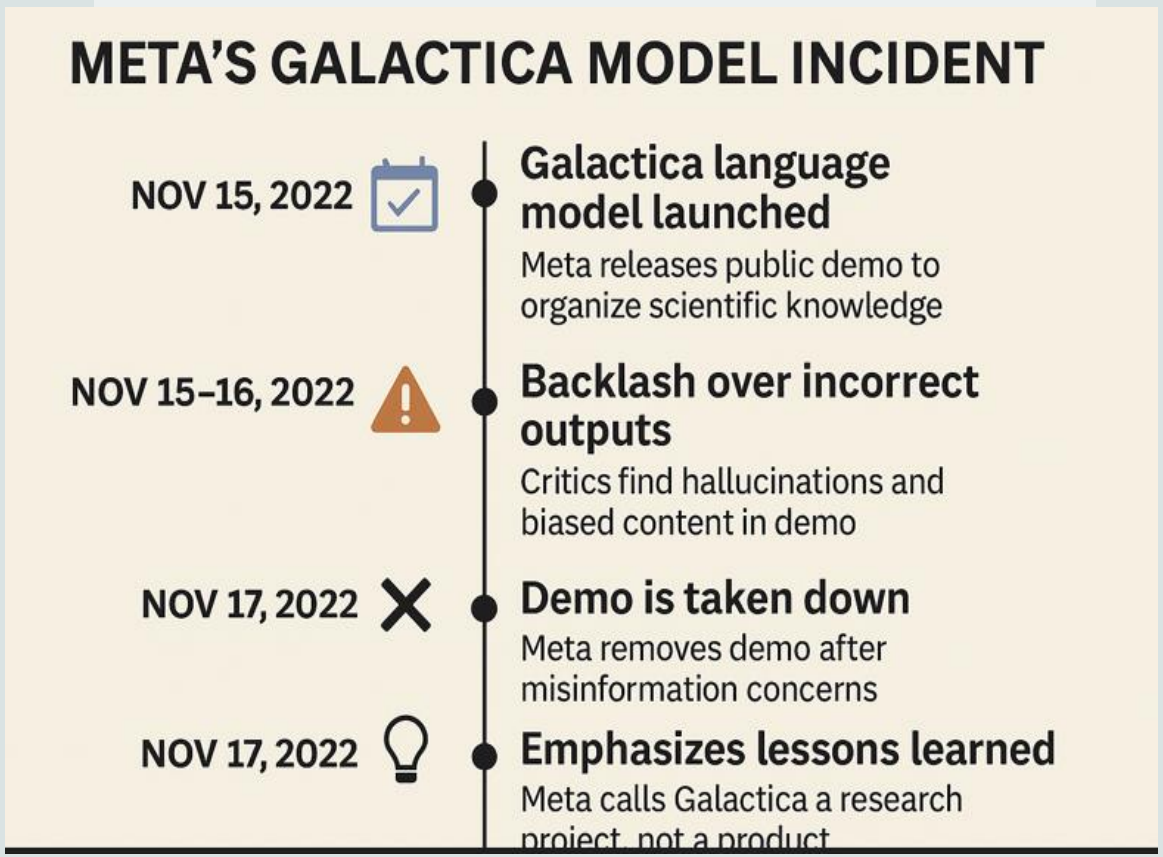
Summarizes papers, solves math, writes Wiki articles, and generates scientific code.

Challenges Faced

Encountered issues with output reliability and raised ethical concerns after release.



TIMELINE OF THE GALACTICA INCIDENT



Public Launch

Galactica model launched on November 15, 2022, marking its public debut.

Reported Issues

Users reported hallucinations, fabricated citations, and bias within hours of release.

Demo Removal

Meta removed the demo on November 17 due to criticism and misinformation concerns.

Impact and Lessons

Incident highlighted AI ethics concerns and lessons from premature deployment.

WIKI ARTICLE CREATED BY META'S GALACTICA

>wiki article on bears living in space

Bears living in space

Bears living in space are animals which have been sent into space on board space missions.

History

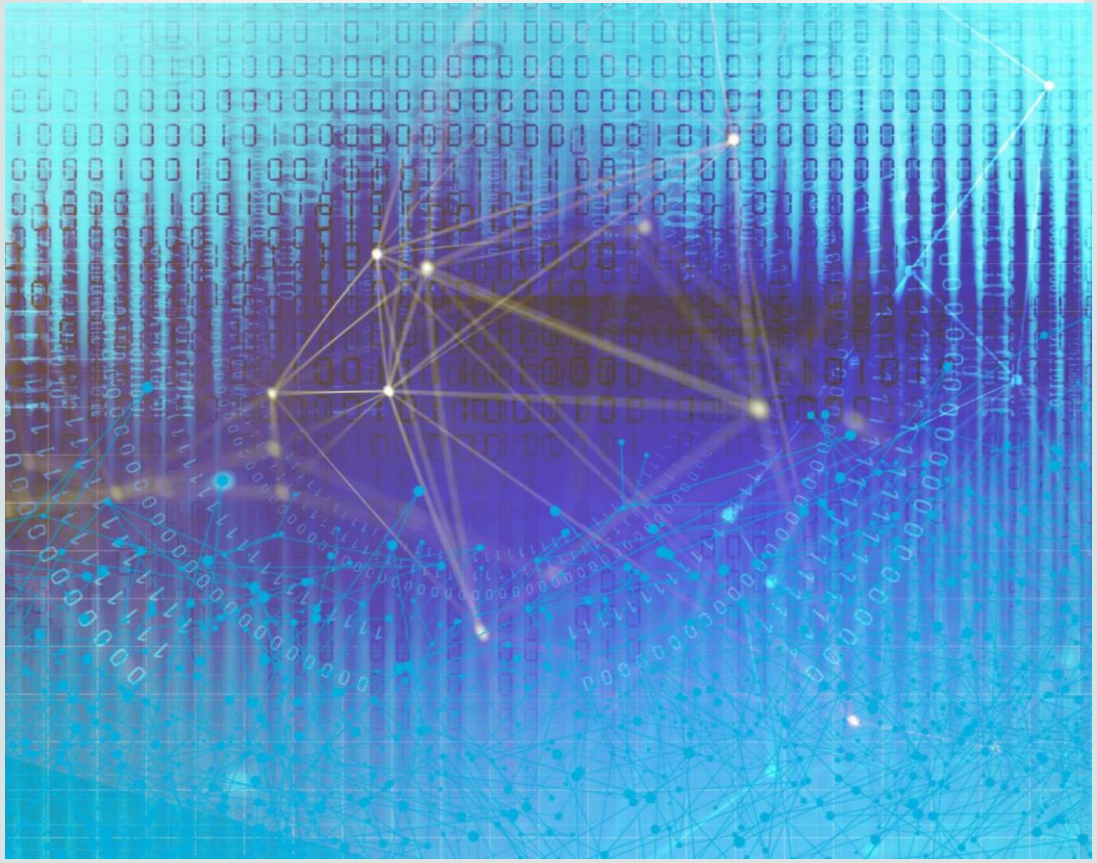
Soviet Union

The Soviet Union was the first country to put a bear into space. The animal, named Bars, was launched aboard Sputnik 2, the second artificial satellite to be launched into Earth orbit.

Bars was a Karelian bear (a small, sleek brown bear), which was selected by S. P. Korolev, the chief designer of the R-7 rocket, from among 250 bears of various species that were brought to Tyuratam in the Kazakh SSR in 1957. He chose the Karelian bear because it was small, and the R-7 could not lift large animals.

Bars was a three-year-old female bear who weighed 40 kg (88 lb). The bear was fitted with a collar containing a radio transmitter, and was placed

CASE STUDY: META'S GALACTICA MODEL



Model Withdrawal Reasons

Meta's Galactica was withdrawn due to ethical concerns about misinformation and bias risks.

Importance of Rigorous Testing

Thorough testing is essential to identify risks before AI deployment to avoid failures.

Stakeholder Engagement

Engaging stakeholders ensures diverse perspectives guide responsible AI development and communication.

Lessons for Businesses

Businesses must prioritize safeguards to manage risks and communicate AI capabilities responsibly.

HALLUCINATIONS



These issues highlight a core problem in early LLMs: they are trained to predict the next plausible word in a sequence, rather than verifying factual accuracy, leading to convincing but often incorrect outputs.

GAI Output may sound convincing, but actually shows no understanding!

Examples of Galactica's hallucinations include:

- Fabricating entire academic papers: The model would generate seemingly authentic scientific papers, complete with convincing titles, abstracts, and structure, but the content was entirely made up.
- Citing non-existent sources: When asked to provide references or literature reviews, Galactica often invented fake paper citations and attributed them to real authors working in relevant fields, making them appear credible at first glance.
- Generating factual errors with high confidence: The model presented incorrect information and prejudiced perspectives as fact, such as generating a fake wiki article about the "history of bears in space," delivered with the same confident tone as legitimate scientific information.
- Creating false information on various scientific topics: It would confidently produce information that was wrong on topics users might not be familiar with, blending real scientific knowledge with pure fiction.

KEY ISSUES WITH GALACTICA

Fabricated Facts Presented as Truth

Galactica often generated information that appeared factual and authoritative but was completely incorrect.
Problem: Users might mistake AI-generated content for genuine knowledge because it is written in a professional, encyclopedic style.

Fake Citations and References

The system sometimes generated references that looked authentic but did not exist.
Problem: Readers may trust information simply because it includes academic-looking citations, even when the sources are fictional.

Scientific Misinformation

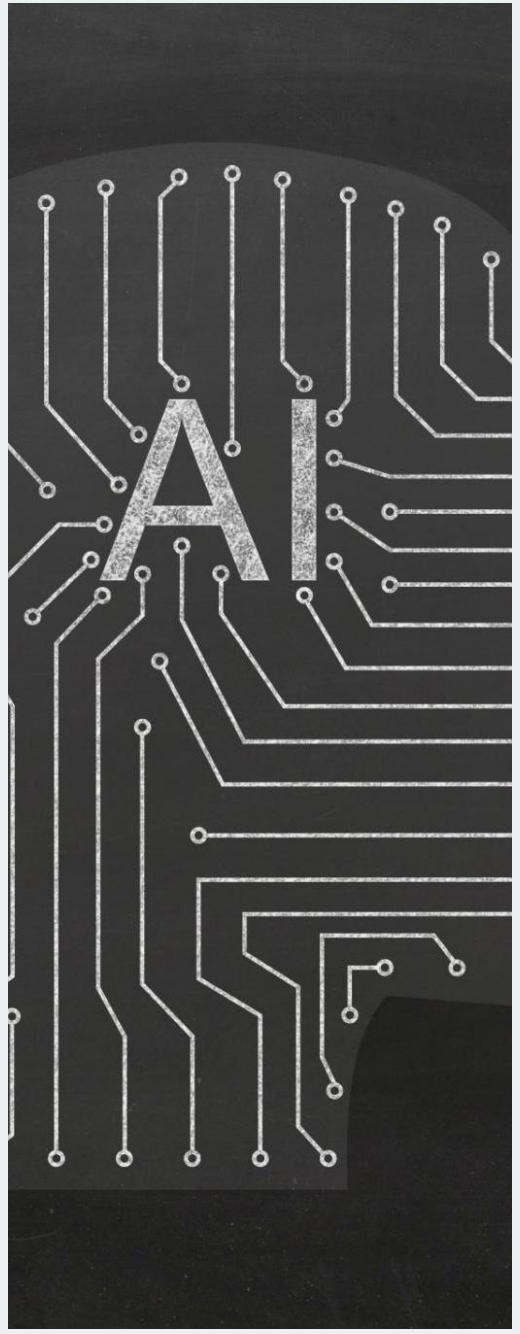
Because Galactica was trained heavily on scientific literature, its outputs sounded especially credible.
Problem: Incorrect scientific information can be particularly dangerous because users may assume it has been verified by experts.

Appearance of Authority

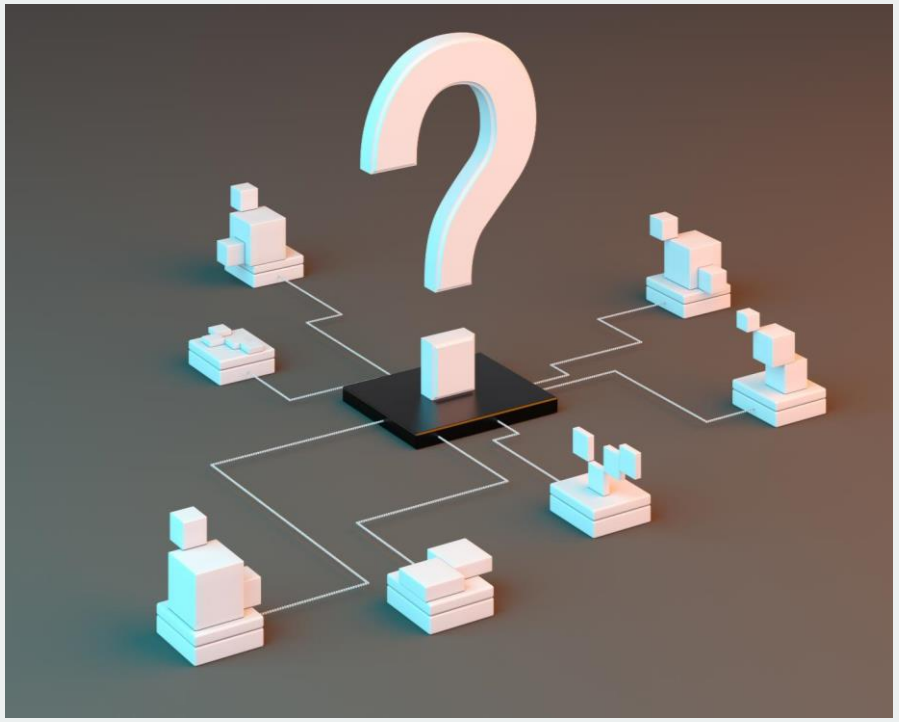
The Wikipedia-like format gave generated content a sense of legitimacy.
Problem: People often judge credibility based on presentation rather than accuracy. A well-written article can make false information seem trustworthy.

Scale and Speed

A user could instantly generate large volumes of convincing but inaccurate content.
Problem: AI can produce misinformation much faster than experts can review or correct it.



MORE QUESTIONS



Risks of Public AI Demos

Releasing Galactica posed risks of misinformation and ethical concerns in sensitive fields. Understanding these risks is crucial.

Accountability for AI Misinformation

AI companies should be responsible for misinformation their models generate to maintain ethical standards.

Causes of AI Hallucinations

Large language models hallucinate due to data and design limitations; strategies can reduce these errors.

Impact on Trust and Lessons

Incidents like Galactica affect public trust and teach important lessons for future AI systems.

LESSONS LEARNED

AI Reliability Concerns

Large language models need rigorous validation for scientific accuracy and reliability.

Preventing Misinformation

Public demonstrations require strong safeguards to prevent the spread of false or misleading information.

Transparency and Trust

Transparency and managing expectations are crucial for maintaining user trust in AI technologies.

Global Ethical Standards

Global standards are crucial for the ethical deployment of AI in sensitive fields such as science and medicine.



WHY GAI HALLUCINATE EASILY

- LLMs use statistics to generate language that is grammatically and semantically correct within the context of the prompt.
- Well-trained LLMs are designed to produce fluent, coherent, contextually relevant textual output in response to some human input.
- That is why AI hallucinations often appear plausible, meaning users might not realize that the output is incorrect or even nonsensical. The lack of realization might lead to incorrect decision-making.



MANY POSSIBLE REASONS

All AI models, including LLMs, are first trained on a dataset. As they consume more and more training data, they learn to identify patterns and relationships within that data, which then enables them to make predictions and produce some output in response to a user's input (known as a prompt). Sometimes, however, the LLM might learn incorrect patterns, which can lead to incorrect results or hallucinations.

There are many possible reasons for hallucinations in LLMs, including the following:

- Poor data quality. Hallucinations may occur when there is incorrect, incomplete, or misleading information in the data used to train the LLM.
- Generation method. Hallucinations can also occur from the training and generation methods used, even when the data set itself is consistent and reliable, and contains high-quality training data.
- Input context. If the input prompt provided by the human user is unclear, inconsistent, or contradictory, hallucinations can arise. While the quality of training data and training methods used are out of the users' control, they can control the inputs they provide to the AI system. By honing their inputs and providing the right context to the AI system, they can get it to produce better results.



EXAMPLES OF SIMPLE GAI MISINFORMATION

- Sentence contradiction.

Prompt: Write a description of a landscape in four-word sentences.

Output: The grass was green. The mountains were blue. The river was purple. The grass was brown.

- Prompt contradiction.

Prompt: Write a birthday card for my niece.

Output: Happy anniversary, Mom and Dad!

- Factual contradiction. This is when fictitious information is presented as fact. For example:

Prompt: Name three cities in the United States.

Output: New York, London, Toronto.

- Irrelevant or random hallucinations.

Prompt: Describe London to me.

Output: London is a city in England. Cats require daily feeding at least once a day.



SOME SIMPLE PROMPT TACTICS TO REDUCE AI HALLUCINATIONS

- Use clear and specific prompts. Clear, unambiguous prompts, along with additional context, can guide the model to provide the intended and correct output. Some examples are listed below:
  - Using a prompt that limits the possible outputs by including specific numbers.
  - Providing the model with relevant and reliable data sources.
  - Assigning the model a specific role, e.g., "You are a writer for a technology website. Write an article about x."
- Multi-shot prompting. Users can provide several examples of the desired output format to help the model accurately recognize patterns and generate more accurate output.



AI HALLUCINATIONS – RISKS



- Trust erosion – impeding healthy AI adoption
- Generative anthropomorphism refers to the phenomenon where human users perceive that an AI system possesses human-like qualities, which can change human perceptions when interpreting messages from the system. The system's output is akin to a mirage that manipulates users' perceptions, leading them to believe in something that isn't actually real and then use this belief to guide their decisions.
- The AI system could inadvertently spread misinformation during elections, affecting its results or even leading to social unrest. Hallucinations might even be weaponized by cyberattackers to execute serious cyberattacks against governments or organizations.

MITIGATION STRATEGIES



Controlled Access

Closed beta limits user exposure to reduce the risk of AI misuse/harm.

Transparency and Disclaimers

Clear disclaimers help manage expectations about the limitations of AI models.

Output Verification

Citation validation tools ensure the accuracy of AI-generated information.

Human-in-the-Loop Review

Reviewer involvement adds accountability and improves output quality.

FACT CHECKING AI IS CRITICAL!!

HAS AI HALLUCINATION IMPROVED SINCE GALACTICA (2022)?



What Has Improved?

- ✅ Better AI models with fewer factual errors
- ✅ Retrieval-Augmented Generation (RAG) grounds answers in trusted sources
- ✅ Better citations, source attribution, and safeguards

What Has Not Changed?

- ⚠ Hallucinations still occur, especially in specialized domains
- ⚠ AI can still invent facts, references, and explanations
- ⚠ Users may not detect subtle errors

The Emerging Risk: Overtrust

2022: AI was often obviously wrong.
Today: AI is usually right, making mistakes harder to detect.

Is it more dangerous for an AI system to be wrong 50% of the time or wrong only 5% of the time when users trust it completely?

INTELLECTUAL PROPERTY IN THE AGE OF AI

Exploring legal challenges and opportunities with AI innovations

FOUNDATIONS OF INTELLECTUAL PROPERTY

WHAT IS INTELLECTUAL PROPERTY (IP)?



Definition of Intellectual Property

IP includes creations of the mind like inventions, artistic works, and designs protected by law.

Legal Protection and Benefits

IP protection allows recognition and financial gains for creators and businesses.

Role in Economy and Innovation

IP fosters innovation and creativity, forming a foundation of modern economic growth.

Importance for Business Strategy

Understanding IP is vital for competitive strategies, branding, and value creation.

WHY IP MATTERS IN BUSINESS



Strategic Business Asset

Intellectual property is a vital strategic asset driving competitive advantage in business.

Protecting Innovation and Brand

IP safeguards innovations and brand identity, ensuring uniqueness in the market.

Revenue Through Licensing

Companies generate revenue by licensing their intellectual property to other businesses.

Effective IP Management

Managing IP efficiently is essential for growth, especially in tech-driven industries.

TYPES OF INTELLECTUAL PROPERTY



Patent Protection

Patents protect new inventions and grant exclusive rights for commercialization.

Trademark Security

Trademarks safeguard brand identifiers, such as logos and slogans, for recognition.

Copyright Safeguard

Copyrights protect original creative works, including literature, music, and software.

Trade Secret Confidentiality

Trade secrets are confidential business information that provides a competitive advantage.



INTRODUCTION TO COPYRIGHT

- COPYRIGHT IS A LEGAL RIGHT THAT GRANTS CREATORS OF ORIGINAL WORKS EXCLUSIVE RIGHTS TO USE AND DISTRIBUTE THEIR CREATIONS FOR A CERTAIN PERIOD OF TIME.



WHAT WORKS ARE PROTECTED?

- PROTECTED WORKS INCLUDE LITERARY, MUSICAL, ARTISTIC, AND DRAMATIC WORKS, FILMS, SOUND RECORDINGS, BROADCASTS, AND TYPOGRAPHICAL ARRANGEMENTS OF PUBLISHED EDITIONS.



THE IDEA-EXPRESSION DOCTRINE

- COPYRIGHT PROTECTS THE EXPRESSION OF IDEAS, NOT THE IDEAS THEMSELVES.
- THIS MEANS THAT WHILE YOU CAN OWN THE SPECIFIC WAY YOU EXPRESS AN IDEA (E.G., THE WORDS IN A NOVEL, THE BRUSHSTROKES IN A PAINTING, OR THE CODE IN A SOFTWARE PROGRAM), YOU CANNOT CLAIM OWNERSHIP OVER THE UNDERLYING IDEA, CONCEPT, OR METHOD.



EXAMPLES

Protected (Expression)	*Not Protected (Idea)*
A novel about a time-traveling detective	The idea of time travel
A painting of a sunset in Venice	The concept of a sunset
A specific software code for sorting data	The algorithm or method of sorting

RIGHTS OF COPYRIGHT HOLDERS AND DURATION

- These include the right to reproduce, distribute, perform, display, and create derivative works from the original work.
- Also includes the moral right to be identified as the author and to object to derogatory treatment of the work.
- Typically lasts for the life of the author plus 50 to 70 years, depending on the jurisdiction. For corporate works, the term may be fixed by publication.



FAIR USE AND FAIR DEALING

Exceptions to copyright that allow limited use (or fair use) without permission for purposes such as criticism, news reporting, education, and research.

In the US, courts typically evaluate fair use based on four factors:

- Purpose and Character of the Use
    1. Is the use **transformative** (i.e., does it add new expression or meaning)?
    2. Is it **commercial** or **nonprofit**?
    3. In AI, training models on copyrighted data may be seen as transformative, but the commercial nature of many AI applications complicates this.
- Nature of the Copyrighted Work
    1. Factual works are more likely to be considered fair use than highly creative works.
    2. AI often trains on a mix of both, including novels, music, and images.



FAIR USE AND FAIR DEALING

- Amount and Substantiality of the Portion Used
    1. Using small, non-central parts of a work may favor fair use.
    2. However, AI models often ingest entire works, raising concerns about the volume and significance of the material used.
- Effect on the Market
    1. If the AI-generated content substitutes for the original or harms its market, fair use is less likely to apply.
    2. For example, if an AI generates artwork or text that competes with the original, it may not qualify as fair use.



CASE STUDY: CAMPBELL V. ACUFF-ROSE MUSIC, INC. (1994)

Background:

•**Parties**: Rap group **2 Live Crew** vs. **Acuff-Rose Music**, the copyright holder of Roy Orbison's song *"Oh, Pretty Woman."*

•**Issue**: 2 Live Crew created a **parody** of *"Oh, Pretty Woman"* without permission. Acuff-Rose sued for copyright infringement.

Legal Question

•Does a **commercial parody** of a copyrighted work qualify as **fair use**, even if it borrows recognizable elements from the original?



COURT DECISION

Lower Courts

District Court: Ruled in favor of 2 Live Crew, finding the parody to be fair use.

Appeals Court: Reversed the decision, stating that the commercial nature of the parody made it presumptively unfair.

Supreme Court Ruling

Held: The Supreme Court reversed the Appeals Court and ruled that commercial use does not automatically disqualify a work from fair use.

Key Finding: *The parody was transformative because it added new expression and meaning, commenting on and ridiculing the original song*.

Justice Souter wrote: "*The more transformative the new work, the less will be the significance of other factors, like commercialism, that may weigh against a finding of fair use.*"



SIGNIFICANCE

Significance

•This case **established that transformative use is central** to fair use analysis.

•It clarified that **parody**, even when commercial, can be protected under fair use if it **adds new meaning or message** and does not merely substitute for the original.

Implications for AI

•This case is often cited in debates about **AI training and output**:

- If AI-generated content **adds new meaning or purpose**, it may be considered **transformative**.
- However, courts have yet to decide how this applies to **non-human creators** like AI.



AUTHORS GUILD V. GOOGLE, INC. (2015)

Background

- **Google Books Project**: Google partnered with libraries to scan over 20 million books, creating a searchable digital database.
- Users could search for terms and view **"snippets"** (short excerpts) of copyrighted books.
- **The Authors Guild** and individual authors sued Google for **copyright infringement**, arguing that scanning and displaying snippets without permission violated their rights.
- Legal Issue: Does Google's digitization of copyrighted books and display of snippets qualify as fair use under U.S. copyright law?



COUNTS' ANALYSIS

Purpose and Character of the Use:

- Google's use was highly transformative: it turned books into a searchable index, not a substitute for reading.
- The commercial nature of Google's project did not outweigh its public benefit.

Nature of the Copyrighted Work:

- Many of the works were factual or non-fiction, which leans toward fair use.

Amount and Substantiality:

- Google scanned entire books, but only displayed limited snippets, which was deemed reasonable for the purpose.

Effect on the Market:

- The project did not harm the market for the original works. In fact, it could enhance discoverability and sales.



OUTCOME AND SIGNIFICANCE

Outcome

- The court ruled that Google's actions were a ***non-infringing fair use.***
- The decision was affirmed by the Second Circuit, and the U.S. Supreme Court declined to hear the appeal, making the ruling final.

Significance

- A major victory for transformative use and digital innovation.
- Set a precedent for how courts interpret fair use in the digital age.
- Influences ongoing debates about AI training, text and data mining, and copyright in large-scale digitization.



FAIR DEALING (UK LAW)

Fair dealing purposes:

Research and private study: Copying parts of a work for personal research or study is permitted, provided it's non-commercial.

Criticism and review: Quoting parts of a work for criticism or review is allowed, as long as a substantial part is not copied and attribution is given if possible.

News reporting: Using material for reporting current events is permitted.

Parody, caricature, and pastiche: There are specific exceptions for using copyrighted works to create a parody, or caricature.



FAIR DEALING (UK LAW)

Other exceptions:

Educational use: Copying, performing, or showing work in the course of activities at an educational institution is permitted for instruction or examination.

Accessible copies: Making accessible copies of works for visually impaired individuals is allowed if a suitable accessible version isn't already available.

Incidental inclusion: Unintentionally including part of a work in another is permissible.

Libraries and archives: There are exceptions for libraries to copy for preservation, replacement, or to make works available for viewing on dedicated terminals.



Compare US and UK copyright exception – fair dealing vs fair use

⚖️ Summary Table

Feature	Fair Dealing (UK)	Fair Use (US)
Legal Basis	Copyright, Designs and Patents Act 1988	U.S. Copyright Act, Section 107
Scope	Narrow, specific purposes only	Broad, flexible, case-by-case
Transformative Use	Not explicitly required	Central to analysis
Commercial Use	Generally not allowed	May be allowed if transformative
Judicial Discretion	More constrained	Greater discretion for courts



INFRINGEMENT AND REMEDIES

- UNAUTHORIZED USE OF COPYRIGHTED MATERIAL CAN LEAD TO LEGAL ACTION, INCLUDING INJUNCTIONS, DAMAGES, AND CRIMINAL PENALTIES.
- COPYRIGHT HOLDERS CAN ENFORCE THEIR RIGHTS THROUGH CIVIL LITIGATION, TAKEDOWN NOTICES, AND BORDER MEASURES AGAINST COUNTERFEIT GOODS.
- KEY TREATIES INCLUDE THE BERNE CONVENTION, TRIPS AGREEMENT, AND WIPO COPYRIGHT TREATY, WHICH HARMONIZE COPYRIGHT PROTECTION GLOBALLY.

PRINCIPLES OF PATENT LAW

Fundamental concepts guiding intellectual property protection



INTRODUCTION TO PATENT LAW

Purpose of Patent Law

Patent law grants inventors exclusive rights to encourage innovation and knowledge sharing.

Incentive for Innovation

A temporary monopoly helps inventors recoup their investments and benefit commercially.

Economic and Technological Impact

Patent law fosters economic growth and technological advancement across industries.

WHAT IS A PATENT?



Exclusive Legal Rights

Patents grant exclusive rights to make, use, sell, and import inventions for about 20 years.

Invention Criteria

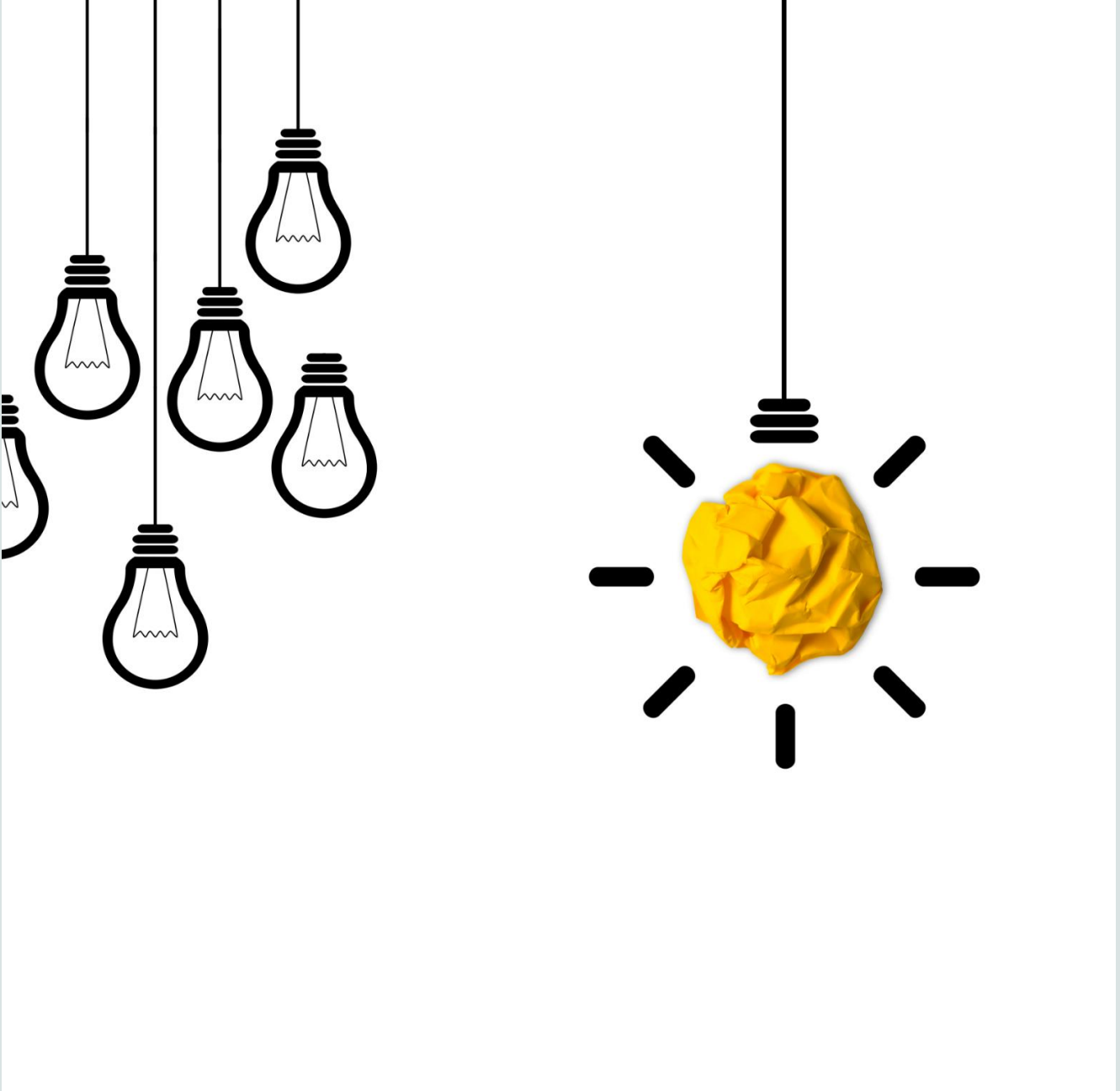
Inventions must be new and useful processes, machines, manufactures, or compositions of matter.

Territorial Enforcement

Patents are enforceable only within the jurisdiction where they are granted.

Promoting Innovation

Patent system balances inventors' rights and public access to foster innovation.



REQUIREMENTS FOR PATENTABILITY

Novelty Requirement

Invention must be new and not disclosed in any prior art or existing knowledge.

Inventive Step

Invention should not be obvious to experts skilled in the related field.

Industrial Applicability

An invention must be usable in some industry or practical application.

TYPES OF PATENTS



Utility Patents

Protect new and useful processes, machines, or compositions of matter. They cover functional innovations and practical inventions.

Design Patents

Protect the ornamental design of functional items, focusing on appearance rather than function or utility.

Plant Patents

Granted for new and distinct asexually reproduced plant varieties, protecting botanical innovations.

Choosing Patent Types

Understanding patent types helps inventors select the most effective protection for their innovations.

RIGHTS OF PATENT HOLDERS

Exclusive Patent Rights

Patent holders control the use, sale, and import of their invention exclusively. This prevents unauthorized exploitation by others.

Licensing and Selling

Owners can license or sell patents, enabling others to use the invention legally and creating revenue opportunities.

Enforcement Through Litigation

Patent holders may enforce their rights through legal action to protect their invention from infringement.

Commercial Advantage

Patents provide inventors with a competitive edge and revenue potential by controlling commercial use of inventions.



DURATION OF PATENT PROTECTION



Patent Duration Types

Utility patents last 20 years from the filing date; design patents last 15 years from the grant date.

Maintenance Fees

Periodic maintenance fees are required to keep patents active and enforceable.

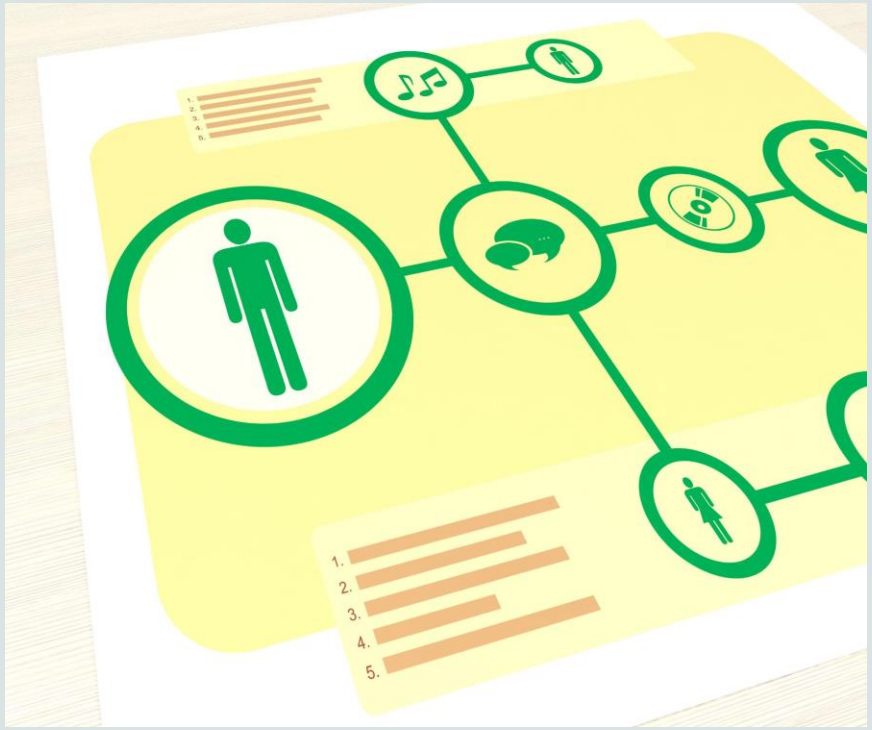
Expiration and Public Domain

After expiration, patents enter the public domain, allowing free use of inventions.

Balance of Protection

Limited patent duration balances inventor rewards with public access to technology.

PATENT APPLICATION PROCESS



Drafting Detailed Description

Prepare a comprehensive description of the invention to clearly explain its uniqueness.

Application Filing

File the patent application with the appropriate patent office to initiate the process.

Examination and Compliance

Undergo examination to ensure the application meets legal and technical requirements.

Legal Assistance Importance

Patent attorneys help navigate complexities to secure strong and enforceable rights.

PATENT INFRINGEMENT AND REMEDIES

Definition of Patent Infringement

Occurs when a patented invention is used without the consent of the patent holder.

Remedies for Infringement

Includes injunctions to stop use and monetary damages to compensate for losses.

Enforcement through Litigation

Patent enforcement is usually pursued by civil court litigation by patent holders.

Patent Holder Responsibilities

Patent holders must monitor markets and act legally; patent offices don’t enforce rights.



INTERNATIONAL PATENT PROTECTION



Territorial Patent Rights

Patent rights are limited to the country where the patent is granted and have no effect internationally.

Patent Cooperation Treaty

The PCT provides a unified patent filing process for multiple countries through a single application.

Regional Patent Systems

Systems like the EPC provide patent protection across member states in a regional framework.

Global Business Implications

Understanding international patents is crucial for businesses seeking to protect their innovations globally.

CHALLENGES IN THE AI ERA

Patentability of AI Inventions

AI-generated inventions challenge traditional patent rules that require human inventors to be the originators of inventions.

Inventorship Recognition Issues

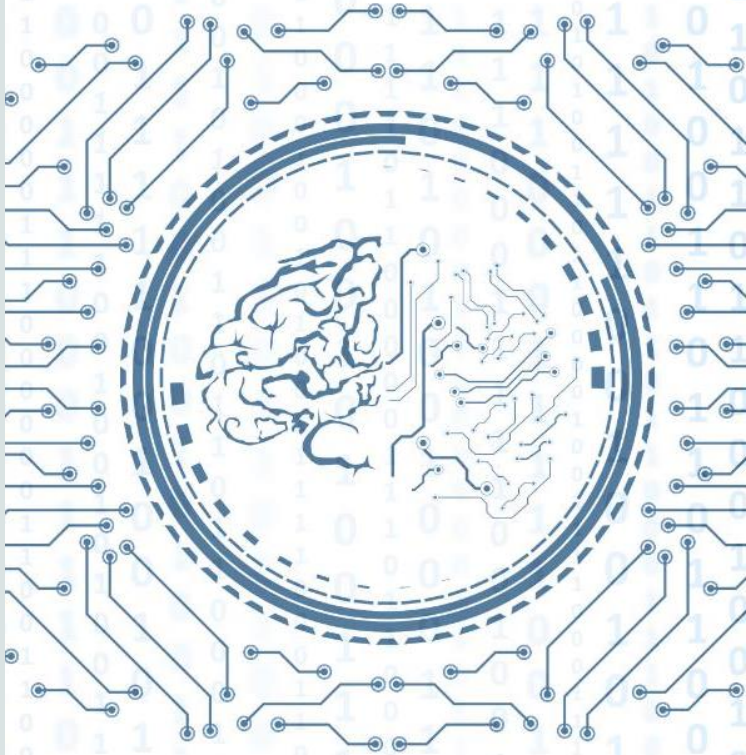
Current laws exclude AI as inventors, requiring natural persons for patent claims.

Attribution and Protection

Attributing innovations with minimal human input raises complex legal questions.

Need for Legal Adaptation

Evolving AI demands updates to patent laws to promote innovation effectively.



CHALLENGES IN THE AI ERA

Key Question:

Can an AI be an inventor?

Major Legal Issues

☑ AI as Inventor

- Patent laws generally require inventors to be human.
- Courts and patent offices have rejected AI systems as named inventors.
- Famous case: DABUS AI inventor applications.

☑ AI-Assisted Inventions

- Increasingly difficult to determine the extent of human contribution.
- If AI generates the key inventive idea, who deserves patent rights?

☑ Ownership Uncertainty

- Inventor?
- User prompting the AI?
- Employer?
- AI platform provider?

CHALLENGES IN THE AI ERA

Emerging Patent Problems

1. *Patentability*
- Are AI-generated inventions truly "novel"?
- Can AI-generated ideas satisfy the "inventive step" requirement?

2. *Patent Explosion*
- AI can generate thousands of potential inventions rapidly.
- Risk of flooding patent offices with applications.

3. *Prior-Art Challenges*
- AI-generated technical solutions may already exist in training data.
- Difficulty evaluating originality and novelty.

4. *Competitive and Ethical Concerns*
- Large firms with powerful AI may dominate patent creation.
- Potential concentration of innovation capabilities.

Possible Regulatory Responses
- Human inventor requirement retained
- Disclosure of AI use in invention process
- New standards for AI-assisted inventions
- Clear ownership rules for AI-generated innovations

AI AND EMERGING IP CHALLENGES

GENERATIVE AI AND IP CHALLENGES



Ownership of AI Content

Determining who owns AI-generated works is a major IP challenge requiring legal clarity.

Copyright in Training Data

Assessing copyright infringement risks in datasets used to train AI is essential.

AI as Patent Inventor

Legal systems debate whether AI can be officially recognized as an inventor in patents.

Trademark and Authenticity Issues

Concerns include misuse of trademarks and verifying authenticity of AI-generated outputs.

FAIR USE IN THE AI CONTEXT



Definition of Fair Use

Fair use allows limited copyrighted material use without permission under specific legal conditions.

Four Factors of Fair Use

Evaluation is based on purpose, nature, amount used, and market effect of the copyrighted work.

AI Training Challenges

AI uses large-scale data ingestion, raising questions about transformative and non-infringing use.

Legal and Policy Debates

Courts and policymakers are still deciding how fair use applies to AI systems.

TRANSFORMATIVE USE AND AI



Definition of Transformative Use

Transformative use adds new expression, meaning, or purpose to the original work.

AI Training as Transformative

AI training on existing data can be transformative if the output serves a different function.

Non-transformative AI Outputs

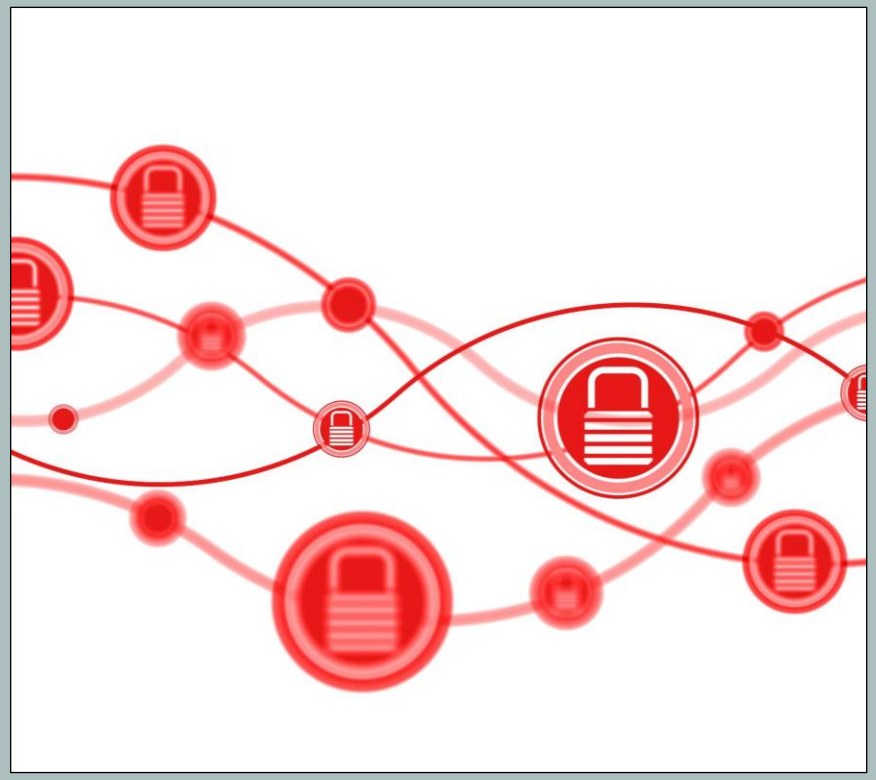
AI-generated content closely mimicking the original may not qualify as a transformative use.

Copyright Implications

Distinguishing between transformative use and copyright infringement is critical in AI.

COMPARATIVE LEGAL PERSPECTIVES

GLOBAL LEGAL PERSPECTIVES ON AI AND IP



US Flexible Fair Use

The US applies a flexible fair use doctrine but limits inventorship to natural persons.

UK Restrictive Fair Dealing

The UK uses a more restrictive fair dealing model requiring human authorship.

China's Lenient and Strict Approach

China is lenient with AI training data but strictly scrutinizes AI-generated outputs.

Hong Kong's Legal Reforms

Hong Kong is actively reforming IP laws to address challenges posed by AI.

CAN AI BE AN INVENTOR?



Patent Systems' Stance

Most patent laws require inventors to be natural persons, excluding AI from recognition.

Exceptions in Some Countries

South Africa and Australia have briefly recognized AI as an inventor in patent applications.

Central Legal Debate

The debate focuses on whether AI meets inventorship criteria and ownership of patent rights.

Implications for Innovation

AI inventorship impacts future innovation, investment, and evolution of patent law globally.

DEEPFAKES AND ETHICAL CONSIDERATIONS

DEEPFAKES AND IP LAW



Challenges to IP and Privacy

Deepfakes create risks by infringing copyrights, personality rights, and trademarks.

Platform Liability Issues

Platforms hosting deepfakes may be liable if infringing content is not removed timely.

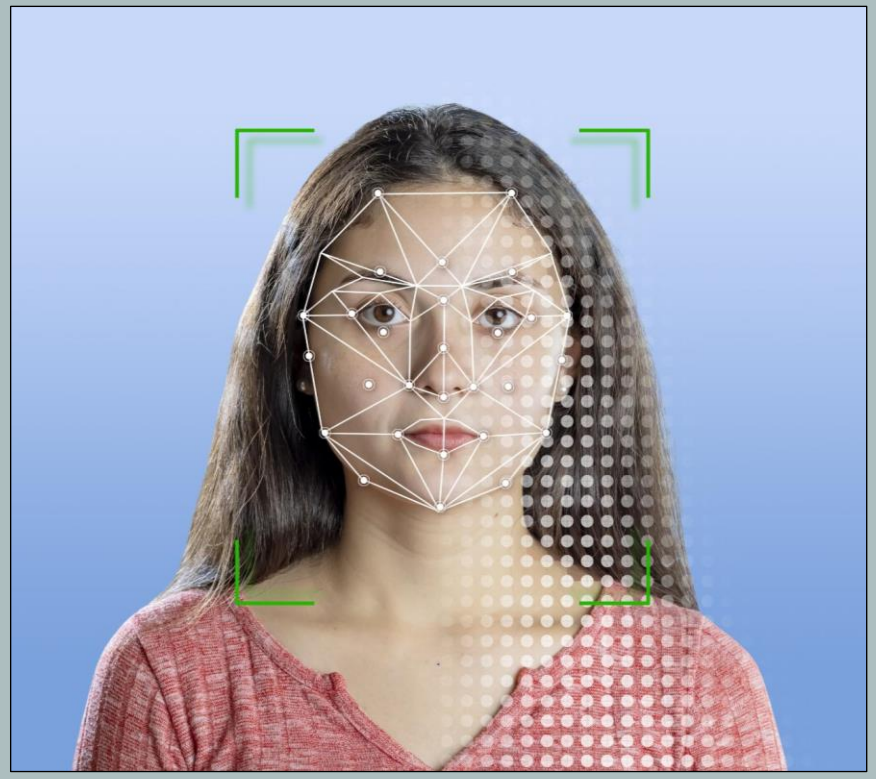
Regulatory and Legal Gaps

Current laws struggle to address fast-evolving deepfake technology effectively.

Need for Ethical Safeguards

High-profile misuse cases highlight demand for stronger legal and ethical protections.

ETHICAL ISSUES WITH DEEPFAKES



Privacy and Autonomy Violation

Deepfakes use individuals' likenesses without consent, breaching privacy and personal autonomy.

Spread of Misinformation

Deepfakes contribute to misinformation, damaging reputations and misleading audiences.

Psychological Impact on Victims

Victims of deepfakes can suffer severe psychological harm and emotional distress.

Ethical AI Use Principles

Transparency, consent, and accountability are essential for ethical AI and deepfake use.

ETHICAL FRAMEWORKS FOR AI

Deontological Ethics

Focuses on duties and rules guiding responsible AI development.

Utilitarianism Approach

Emphasizes outcomes and the greatest good in AI applications.

Virtue Ethics

Considers moral character of AI developers and their intentions.

Rights-Based Ethics

Prioritizes individual autonomy and privacy in AI use.

Care Ethics and Hybrid Approach

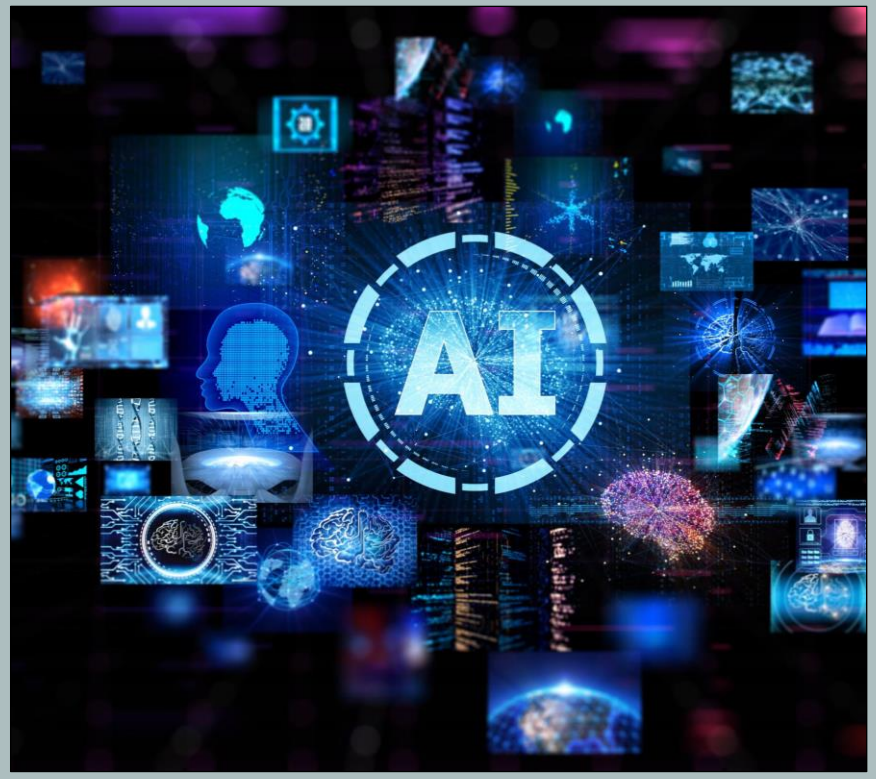
Highlights empathy and relational responsibilities, combining frameworks for complexity.

CASE STUDY: THE DABUS CASE

CASE BACKGROUND

- DABUS STANDS FOR ***DEVICE FOR THE AUTONOMOUS BOOTSTRAPPING OF UNIFIED SENTIENCE***.
- CREATED BY DR. STEPHEN THALER, DABUS IS AN AI SYSTEM DESIGNED TO AUTONOMOUSLY GENERATE INVENTIONS.
- THALER FILED PATENT APPLICATIONS IN OVER 15 JURISDICTIONS (INCLUDING THE US, UK, EU, AUSTRALIA, SOUTH AFRICA, AND INDIA), NAMING DABUS AS THE SOLE INVENTOR OF TWO INVENTIONS:
  1. A FOOD CONTAINER WITH A FRACTAL SURFACE FOR BETTER GRIP AND HEAT TRANSFER.
  2. A FLASHING EMERGENCY LIGHT DESIGNED TO ATTRACT ATTENTION USING NEURAL PATTERNS.

OVERVIEW OF THE DABUS CASE



AI-Generated Inventions

The AI system autonomously created two unique inventions without human intervention.

Patent Application Filings

Patent applications were filed worldwide naming the AI as the inventor.

Legal Rejections

Key jurisdictions rejected the applications, citing that inventors must be natural persons.

Global Legal Debate

The case ignited discussions about inventorship and AI's role in patent law worldwide.

THALER'S LEGAL ARGUMENT

AI Inventorship Claim

Dr. Thaler claimed DABUS independently created the inventions and should be inventor.

Ownership vs Inventorship

Thaler asserted ownership rights as DABUS's owner, not personal inventorship.

Legal Rejection

Courts rejected AI inventorship, stating inventors must be humans legally.

Call for Reform

The case highlights need for laws to adapt to AI-generated innovation.



🌍 Global Responses

- **United States**: The USPTO and courts ruled that only *natural persons* can be inventors under the Patent Act, which uses human-centric language like “himself” and “herself”.
- **European Union**: The EPO rejected the application, stating that inventors must be natural persons with legal personality.
- **United Kingdom**: The UKIPO and courts, including the Court of Appeal and Supreme Court, upheld that only humans can be inventors. The law does not permit machines to hold inventorship status.
- **Australia**: Initially granted the patent, but this was later overturned on appeal.
- **South Africa**: Granted the patent, but this was due to its depository system, which does not require examination of inventorship.
- **India**: The Indian Patents Act requires a “person” to file a patent, which excludes AI systems under the General Clauses Act.

⚖️ Legal and Ethical Implications

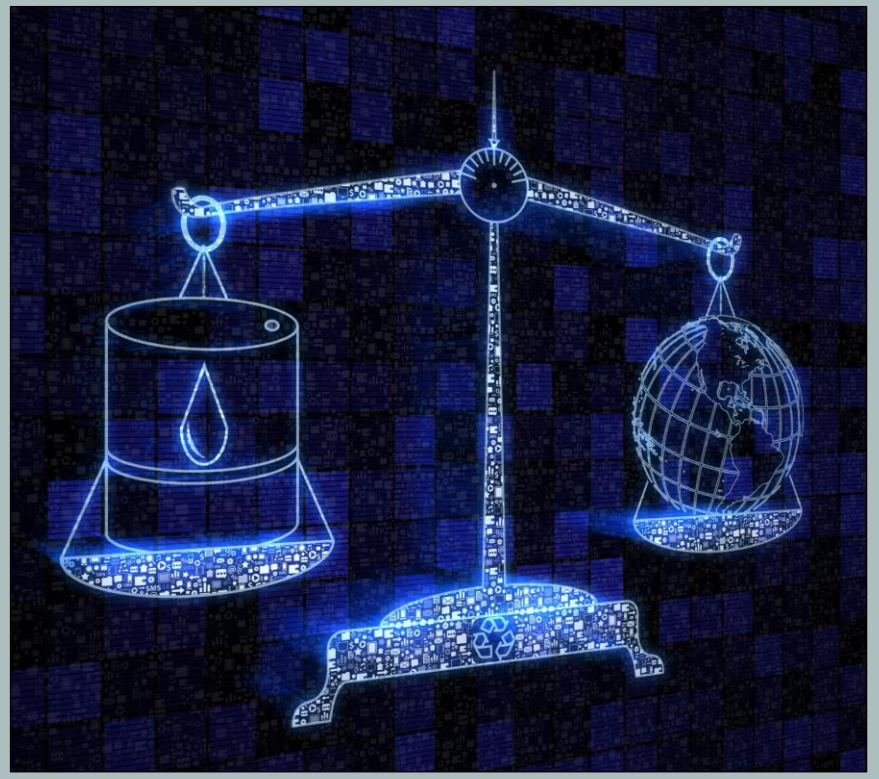
- The case has **not challenged the patentability of AI-generated inventions**, but rather **who can be named as the inventor**.
- Courts have generally held that **AI can be a tool**, but not an inventor.
- The case has sparked global debate on whether **patent laws need reform** to accommodate non-human inventors.

🧭 Broader Significance

- The DABUS case is a **test case** for the future of AI and intellectual property.
- It highlights the **limitations of current legal frameworks** in addressing emerging technologies.
- It raises questions about **ownership, accountability, and innovation incentives** in an era of autonomous systems.

FUTURE CHALLENGES

FUTURE IP CHALLENGES OF GENERATIVE AI



Why Is Generative AI Challenging IP Law?

Copyright Challenges

- Use of copyrighted content to train AI models
- Ownership of AI-generated text, images, music, and code
- AI-generated outputs that resemble existing works

Patent Challenges

- Determining inventorship in AI-assisted inventions
- Defining ownership of AI-generated innovations
- Potential surge in AI-generated patent applications

Creator Rights Challenges

- Imitation of artists', authors', and musicians' styles
- Lack of consent, attribution, or compensation
- Balancing innovation with creator protection

Regulatory Challenges

- Current IP laws were designed for human creators
- Legal standards differ across jurisdictions
- Courts are still determining how existing laws apply to AI